

3.1.3 Operating Condition

The following terms are used in emergency procedures to describe the operating condition of a system, subsystem, assembly or component:

Affected Fails to operate in the normal or usual manner.

Normal Operates in the normal or usual manner.

3.1.4 Urgency of Landing

NOTE The type of emergency and the emergency conditions combined with the pilot's analysis of the condition of the helicopter and his proficiency are of prime importance in determining the urgency of a landing.

The following terms are used to reflect the degree of urgency of an emergency landing:

LAND IMMEDIATELY

The urgency of landing is paramount. Primary consideration is to assure survival of the occupants. Landing in water, trees or other unsafe areas should be considered only as a last resort.

LAND AS SOON AS POSSIBLE

Land without delay at the nearest adequate site (i.e. open field) at which a safe approach and landing can be made.

LAND AS SOON AS PRACTICABLE

The landing site and duration of flight are at the discretion of the pilot. Extended flight beyond the nearest approved landing area where appropriate assistance can be expected is not recommended.

3.2 WARNING AND CAUTION LIGHTS

Most emergency situations will be indicated by either a **red warning light** or a **yellow caution light** coming on.

A **red warning light** on the WARNING PANEL or in pilot's view indicates an emergency condition requiring immediate corrective action. Additional audio signals may be triggered.

A **yellow caution light** on the instrument panel indicates a malfunction or failure conditions which do not require immediate crew action but the possible need for future corrective action.

It is always possible that a warning/caution light will unnecessarily come on. Whenever possible, check the light against its associated instrument to verify that an emergency condition has actually occurred.

Following is an alphabetical listing of the warning and caution light indications with the respective conditions, any further indications, and the emergency procedure.

3.2.1 Warning Light Indications

WARNING LIGHT INDICATIONS

BAT 60

Conditions/Indications

Battery temperature high (above 60°C)

Procedure

• ON GROUND

- | | |
|---------------|-------------------------|
| 1. Battery sw | – Off |
| 2. Engines | – Leave running in idle |

If warning light remains on for more than 5 minutes:

- | | |
|------------|-------------|
| 3. Engines | – Shut down |
|------------|-------------|

CAUTION BATTERY MUST BE INSPECTED OR REPLACED PRIOR TO NEXT FLIGHT.

• IN FLIGHT

- | | |
|--------------------------------|-------|
| 1. Battery sw | – Off |
| 2. LAND AS SOON AS PRACTICABLE | |

CAUTION BATTERY MUST BE INSPECTED OR REPLACED AT DESTINATION PRIOR TO NEXT FLIGHT.

WARNING LIGHT INDICATIONS

BAT 70

Conditions/Indications

Battery overtemperature (above 70°C)

Procedure

• ON GROUND

- | | |
|---------------|-------------|
| 1. Battery sw | – Off |
| 2. Engines | – Shut down |

CAUTION BATTERY MUST BE INSPECTED OR REPLACED PRIOR TO NEXT FLIGHT.

• IN FLIGHT

- | | |
|---------------|-------|
| 1. Battery sw | – Off |
|---------------|-------|

2. LAND AS SOON AS POSSIBLE

After landing:

- | | |
|---------------------|-------------------------|
| 3. Engines | – Leave running in idle |
| 4. Collective lever | – Lock |
| 5. Cyclic stick | – Centered |

NOTE Continue flight only if visual inspection reveals no indication of battery overheating. Leave battery off or disconnect battery.

CAUTION BATTERY MUST BE INSPECTED OR REPLACED AT DESTINATION PRIOR TO NEXT FLIGHT.

WARNING LIGHT INDICATIONS



(Engine 1)

or



(Engine 2)

Conditions/Indications

Overtemperature in engine compartment

Procedure

• ON GROUND

- | | |
|-----------------------------------|--------------------------|
| 1. Both EMERG FUEL VALVE sw | – CLOSE |
| 2. Both FUEL PUMPS/SUPPLY TANK sw | – Off |
| 3. Both power levers | – OFF |
| 4. Battery sw | – Off |
| 5. Passengers | – Alert/Evacuate |
| 6. Fire | – Extinguish if possible |

• IN FLIGHT

- | | |
|--|---------|
| 1. EMERG FUEL VALVE sw (affected engine) | – CLOSE |
| 2. FUEL PUMPS/SUPPLY TANK sw (affected engine) | – Off |
| 3. Power lever (affected engine) | – OFF |
| 4. Passengers | – Alert |
| 5. LAND AS SOON AS POSSIBLE | |

After landing:

- | | |
|---------|--------------------------|
| 6. Fire | – Extinguish if possible |
|---------|--------------------------|

WARNING LIGHT INDICATIONS**FILT 1**

and/or

FILT 2**Conditions/Indications**

Engine fuel filter(s) clogged

Procedure**• ONE WARNING LIGHT ON**

LAND AS SOON AS PRACTICABLE

Be prepared for single engine failure

• BOTH WARNING LIGHTS ON

LAND AS SOON AS POSSIBLE

Be prepared for double engine failure

WARNING LIGHT INDICATIONS

GEN 1

or

GEN 2

Conditions/Indications

Generator failure

- Voltmeter indication normal (27 - 29 V)
- **GEN** warning light (affected generator) on

Generator overvoltage

- Voltmeter indication above normal (30 V or more)
- **GEN** warning light (normal generator) on

NOTE Normal generator is disconnected by reverse current of affected generator.

Procedure

• GENERATOR FAILURE

1. STARTER/GENERATOR sw (affected generator) – Off, then GEN ON

If **GEN** warning light remains on:

2. STARTER/GENERATOR sw (affected generator) – Off
3. CURRENT IND sw – Select normal generator
4. Ammeter and voltmeter – Monitor

• GENERATOR OVERVOLTAGE

1. STARTER/GENERATOR sw (affected generator) – Off
2. **GEN** warning light (affected generator) – On
3. **GEN** warning light (normal generator) – Off
4. CURRENT IND sw – Select normal generator
5. Ammeter and voltmeter – Monitor

NOTE One generator alone will provide sufficient power for normal services. Non-essential DC-bus loads are shed automatically.

WARNING LIGHT INDICATIONS

GEN 1

and

GEN 2

Conditions/Indications

Both generators have failed or are disconnected from the power distribution circuit.

Procedure

1. Each STARTER/GENERATOR sw in turn – Off, then GEN ON

If both **GEN** warning lights remain on:

2. Both STARTER/GENERATOR sw – Off
3. CURRENT IND sw – BUS BAR (BATT)
4. Electrical consumption – Reduce
5. Ammeter and voltmeter – Monitor
6. LAND AS SOON AS PRACTICABLE

Residual Battery Endurance					
Continuous load [A]	15	20	25	30	40
Time [min]	60	45	35	30	22
NOTE Calculations are based on an assumed minimum battery capacity of 15 Ah. Times include 10-min landing light operation and 10-min radio transmitting.					

WARNING A TOTAL ELECTRICAL FAILURE (LOSS OF BOTH GENERATORS AND ONBOARD BATTERY POWER) WILL LIMIT THE FUEL AVAILABLE TO THAT QUANTITY CONTAINED IN THE SUPPLY TANK AT THE TIME OF FAILURE.

WARNING LIGHT INDICATIONS

EFFECTIVITY *Hydraulic Systems without Positive Cross-over Indication*

**HY.
BLOCK**

EFFECTIVITY *Hydraulic Systems with Positive Cross-over Indication*

**HY.
BLOCK**

and

HYD 1

EFFECTIVITY *ALL*

Conditions/Indications

Servo jam in system 1 with automatic cross-over to system 2

- Increased force required to move the affected control axis

Procedure

CAUTION • DO NOT OPERATE HYD TEST SWITCH IN FLIGHT.

- DURING ENGINE SHUTDOWN, BATTERY SHALL REMAIN ON UNTIL ROTOR COMES TO A COMPLETE STOP.

LAND AS SOON AS PRACTICABLE

WARNING LIGHT INDICATIONS

HYD 1

Conditions/Indications

System 1 pressure loss with automatic cross-over to system 2

Procedure

- CAUTION**
- DO NOT OPERATE HYD TEST SWITCH IN FLIGHT.
 - DURING ENGINE SHUTDOWN, BATTERY SHALL REMAIN ON UNTIL ROTOR COMES TO A COMPLETE STOP.

LAND AS SOON AS PRACTICABLE

WARNING LIGHT INDICATIONS**HYD 2****Conditions/Indications**

System 2 pressure loss, system 1 retains power

Procedure

CAUTION • DO NOT OPERATE HYD TEST SWITCH IN FLIGHT.

- AVOID ABRUPT MANEUVERS AND BANK ANGLES GREATER THAN 30°.

LAND AS SOON AS PRACTICABLE

WARNING LIGHT INDICATIONS

LIMIT



MAST MOMENT INDICATOR

Conditions/Indications

MM indicator between end of **yellow arc** and 95% of full deflection

- **LIMIT** warning light on, however, goes off when decreasing mast moment

MM indicator **95% of full deflection** exceeded

- **LIMIT** warning light on and remains on even when decreasing mast moment

Procedure

● BETWEEN END OF YELLOW ARC AND 95% OF FULL DEFLECTION

- | | |
|-------------------------------|------------------|
| 1. Control input | – Reduce |
| 2. LIMIT warning light | – Check goes off |

● 95% OF FULL DEFLECTION

- | | |
|--------------------------------|-----------------------------|
| 1. Control input | – Reduce |
| 2. MM indicator | – Maintain within green arc |
| 3. LAND AS SOON AS PRACTICABLE | |

NOTE Even once the MM indicator exceeds 95% of full deflection (close to white radial) and the **LIMIT** warning light remains on the warning light can be reset by pushing the **TEST** button.

WARNING LIGHT INDICATIONS

LOW FUEL

Conditions/Indications

Failure of both main tank pumps, or
Main tank empty

- Supply tank fuel quantity below 60 kg (75 l)

Procedure

- | | |
|---|------------|
| 1. Supply tank fuel level | – Check |
| 2. Main tank fuel level | – Check |
| 3. Both fuel main pump circuit breakers | – Check in |
| 4. Both fuel main pump switches | – Check on |

If **LOW FUEL** warning light remains on:

5. LAND WITHIN 10 MINUTES

WARNING LIGHT INDICATIONS**MAG
PLUG 1**

or

**MAG
PLUG 2****Conditions/Indications**

Metal particles detected in engine oil

Procedure

WARNING A **MAG PLUG** WARNING LIGHT COMING ON TOGETHER WITH EITHER A TORQUE FLUCTUATION OR LOSS OF OIL PRESSURE SIGNIFIES AN IMPENDING ENGINE FAILURE. PERFORM SINGLE ENGINE EMERGENCY SHUTDOWN OF AFFECTED ENGINE IMMEDIATELY.

• ON GROUND

1. Single engine emergency shutdown – Perform

• IN FLIGHT

1. Single engine emergency shutdown – Perform

2. LAND AS SOON AS PRACTICABLE

If OEI operation is not possible:

3. LAND AS SOON AS POSSIBLE

EFFECTIVITY 0451-9999, and 0001-0450 After SB 60-37

NOTE For procedures applicable to helicopters 0001-0450 Before SB 60-37 see para 3.6.5, Oil Cooler Fan Failure.

WARNING LIGHT INDICATIONS

OIL COOL

Conditions/Indications

Failure of oil cooler fan

Procedure

• AT HOVER (IGE/OGE)

1. Engine and transmission oil temperatures – Monitor

2. LAND AS SOON AS POSSIBLE

• IN FLIGHT

1. Engine and transmission oil temperatures – Monitor

2. Airspeed – Attain as high as possible with minimum power required

3. LAND AS SOON AS PRACTICABLE

If any oil temperature nears limit:

4. LAND AS SOON AS POSSIBLE

NOTE Transmission oil temperature transient range of 105-120 °C has a 10-min limit.

EFFECTIVITY ALL

WARNING LIGHT INDICATIONS

R P M

flashing

Conditions/Indications

N_1 Split

- N_1 indicators show difference of 12% or more
- Audio signal - beeping tone

N_{R0} Low

- N_{R0} 95% or less
- Audio signal - beeping tone

N_{R0} High

- N_{R0} 103% or more
- Audio signal - none

NOTE If a two-stage warning unit is installed, a steady tone will come on with N_{R0} 108% or more.

Procedure

• N_1 S P L I T

- | | |
|---------------------|---------------------------------|
| 1. Collective lever | – Adjust to OEI-limits or below |
| 2. Engine condition | – Analyze |

NOTE RPM warning light may come on during beep trim operation. Try to match N_1 until warning light goes off.

If residual torque is available:

- | | |
|--------------------|-----------------|
| 3. Affected engine | – Leave running |
|--------------------|-----------------|

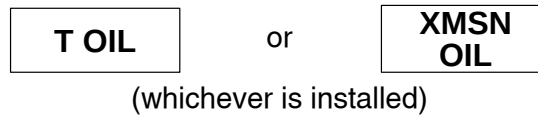
If residual torque is not available:

- | | |
|--------------------------------------|-----------|
| 4. Single engine emergency shut-down | – Perform |
|--------------------------------------|-----------|

• N_{R0} L O W / N_{R0} H I G H

- | | |
|---------------------|--|
| 1. Collective lever | – Adjust as necessary to maintain N_{R0} within normal operating range |
|---------------------|--|

WARNING LIGHT INDICATIONS



Conditions/Indications

Transmission oil **pressure** below minimum

- Transmission oil pressure indication 0.5 kp/cm² or less

Transmission oil **temperature** above maximum

- Transmission oil temperature indication 105 °C or more

Procedure

• OIL PRESSURE BELOW MINIMUM

1. LAND IMMEDIATELY

NOTE Descend with minimum power.

• OIL TEMPERATURE ABOVE MAXIMUM

NOTE Transmission oil temperature transient range of 105 - 120 °C has a 10-min limit.

- | | |
|------------------------------------|---|
| 1. Power | – Reduce |
| 2. Airspeed | – Attain as high as possible with minimum power required |
| 3. LAND AS SOON AS POSSIBLE | |

3.2.2 Caution Light Indications

EFFECTIVITY 0001-0160

CAUTION LIGHT INDICATIONS



EPU ON

EFFECTIVITY 0161-9999

CAUTION LIGHT INDICATIONS



EFFECTIVITY ALL

Conditions/Indications

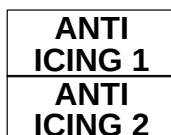
External power is applied to the electrical distribution system

NOTE **EPU ON** caution light going off does not indicate that the EPU cable is disconnected.

Procedure

After EPU starts:

- | | |
|--------------------|----------------|
| 1. EPU cable | – Disconnected |
| 2. EPU access door | – Closed |

EFFECTIVITY 0001-0160**CAUTION LIGHT INDICATIONS****EFFECTIVITY** 0161-9999**CAUTION LIGHT INDICATIONS****EFFECTIVITY** ALL**Conditions/Indications**

Engine anti-icing system in operation

- TOT increases by approximately 15 °C

Procedure

No corrective action necessary.